

Day 20 - Lecture 20.14: Random Variables

 Python 3.10+

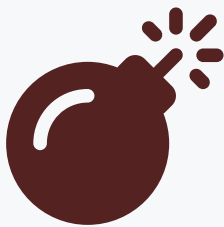
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1. Formal Mathematical Definition

A **Random Variable** is not a traditional algebraic variable; it is a **mathematical function** that maps outcomes from an abstract sample space $\mathcal{S}$ to real numerical values on the real number line $\mathbb{R}$:

$$X : \mathcal{S} \rightarrow \mathbb{R}$$



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2. Taxonomy: Discrete vs. Continuous Random Variables

Property	Discrete Random Variable	Continuous Random Variable
Possible Values	Countable set $x_1, x_2, \dots$	Continuous uncountably infinite interval $[a, b] \subseteq \mathbb{R}$
Probability Function	Probability Mass Function (PMF) $p(x)$	Probability Density Function (PDF) $f(x)$
Single-Point Probability	$P(X = x) \geq 0$	$P(X = x) \equiv 0$
Normalization Law	$\sum_x p(x) = 1.0$	$\int_{-\infty}^{\infty} f(x), dx = 1.0$
Interval Probability	$P(a \leq X \leq b) = \sum_{x=a}^b p(x)$	$P(a \leq X \leq b) = \int_a^b f(x), dx$

3. The Cumulative Distribution Function (CDF)

The **CDF** $F(x)$ represents the probability that random variable X takes a value less than or equal to x , universally defined for both discrete and continuous variables:

$$F(x) = P(X \leq x)$$

Fundamental Properties of CDF:

1. **Monotonic Non-Decreasing:** $x_1 < x_2 \implies F(x_1) \leq F(x_2)$
2. **Boundary Limits:**

$$\lim_{x \rightarrow -\infty} F(x) = 0.0 \quad \text{and} \quad \lim_{x \rightarrow \infty} F(x) = 1.0$$

3. **Relationship to Continuous PDF:**

$$F(x) = \int_{-\infty}^x f(t), dt \iff f(x) = \frac{d}{dx} F(x)$$

4. Python Implementation

```
import numpy as np
import matplotlib.pyplot as plt

# 1. Discrete Random Variable: Number of Heads in 3 Coin Flips
flips_3 = [0, 1, 2, 3]
pmf_discrete = [1/8, 3/8, 3/8, 1/8]
cdf_discrete = np.cumsum(pmf_discrete)

print("Discrete Variable X = Number of Heads:")
for k, p, c in zip(flips_3, pmf_discrete, cdf_discrete):
    print(f"  k = {k}: PMF P(X={k}) = {p:.3f} | CDF F({k}) = {c:.3f}")

# 2. Continuous Random Variable Simulation
samples = np.random.normal(loc=0, scale=1, size=100_000)
p_between = np.mean((samples >= -1.0) & (samples <= 1.0))
print(f"\nContinuous Variable Z ~ N(0, 1): P(-1 <= Z <= 1) = {p_between:.4f} (Exa
```

5. Key Takeaways & Interview Points

- **Point Probability of Continuous RV is Zero:** For a continuous random variable, $P(X = 3.500000...) = 0$. Probability exists only over finite intervals $[a, b]$.
- **CDF is Universal:** Whether discrete, continuous, or mixed, every random variable has a well-defined CDF bounded in $[0, 1]$.